

From Static Screening to Autonomous Synthesis: A Paradigm Shift in Data-Driven Solid-State Battery Discovery

The discovery of solid-state battery (SSB) materials is undergoing a transformative shift from static, descriptor-based screening to dynamic, autonomous scientific reasoning. While early efforts relied on high-throughput filtering of existing databases, recent advancements prioritize physics-informed machine learning interatomic potentials (MLIPs) capable of capturing complex ion transport mechanisms in disordered systems. However, a critical divergence exists: although universal models offer broad applicability, benchmarks indicate that accurate kinetic predictions for dynamically soft chemistries, such as halides, require domain-specific, high-temperature configurational sampling [28] [66]. Furthermore, the field is pivoting from a structure-centric focus to a "synthesis-first" paradigm to bridge the "synthesizability gap," treating executable protocols as primary design variables rather than secondary constraints [214]. This evolution culminates in the emergence of agentic AI systems and self-driving laboratories, which autonomously execute the full discovery cycle from hypothesis generation to experimental validation [287] [145]. By integrating generative design, continual learning, and robotic experimentation, these integrated ecosystems promise to accelerate the realization of stable, high-performance SSBs beyond the limits of human intuition.

Introduction: The Paradigm Shift to Data-Driven Solid-State Battery Discovery

The transition from liquid to solid-state batteries (SSBs) represents a critical frontier in energy storage, driven by the urgent need for higher energy density and improved safety profiles compared to conventional lithium-ion systems. However, the discovery of optimal materials for SSBs is significantly hindered by the vastness of the chemical space and the complexity of physicochemical interactions governing ion transport. Traditional trial-and-error approaches, alongside computationally expensive density functional theory (DFT) methods, are increasingly being supplanted by machine learning (ML) and artificial intelligence (AI) frameworks that accelerate the identification of cathodes, electrolytes, and interfacial solutions. This paradigm shift is evident in the move from static descriptor-based screening to dynamic, physics-informed models capable of capturing ion transport mechanisms and structural disorder. Recent literature reveals a divergence between methods relying on existing databases for compositional screening and those employing generative or first-principles-guided searches for truly novel structures, marking a transformative era in materials discovery [6].

The foundation of this data-driven revolution lies in the curation and expansion of high-quality datasets, particularly those bridging the gap between computed properties and experimental reality. While databases like the Materials Project have long served as repositories for computed properties, recent efforts have focused on creating domain-specific, experimentally validated datasets to train more robust models. For instance, the creation of OBELiX provides a curated database of synthesized solid electrolytes with experimentally measured room-temperature ionic conductivities, addressing the scarcity of reliable experimental data that has previously limited model validation [25]. Similarly, the Energy-GNoME database leverages AI protocols to identify over 33,000 potential energy materials from a pool of 380,000 novel stable crystals, significantly narrowing the search space for thermoelectric and battery applications [274]. In the realm of sodium-ion batteries, large-scale chemically validated databases have been curated from multiple sources to facilitate the discovery of high-voltage cathode-active materials, demonstrating the scalability of integrated data curation strategies [19]. Furthermore, specific datasets targeting migration barriers, such as the literature-derived collection of 619 distinct values, provide the essential ground truth required for training accurate ML models to predict ionic transport properties [223].

Beyond data curation, the methodology for material screening has evolved from simple descriptor-based regression to sophisticated deep learning and generative architectures. Early approaches often relied on facile descriptors derived from elemental properties to predict ionic conductivity, achieving reasonable accuracy in cross-domain classification for superionic conductors [Machine learning-assisted cross-domain prediction of ionic conductivity in sodium and lithium-based superionic conductors using facile descriptors]. However, the limitations of composition-only models, which neglect critical structural factors, have led to the adoption of graph neural networks (GNNs) and universal machine learning interatomic potentials (uMLIPs). Models such as MACE, CHGNet, and MatterSim have demonstrated near-DFT accuracy in predicting energies, forces, and diffusion behaviors at a fraction of the computational cost, enabling the simulation of complex disordered systems and large-scale molecular dynamics [49]. These uMLIPs are particularly crucial for studying high-entropy solid electrolytes and halide systems, where traditional DFT methods are prohibitively expensive due to the vast configurational space [22]. Notably, the dynamic softness of halides poses a stringent test for general-purpose models, prompting the creation of specialized datasets like AQVolt26 that include high-temperature configurational sampling to ensure model robustness under operating conditions [28].

A significant divergence in current research strategies exists between screening known chemical spaces and generating entirely new structures. On one hand, high-throughput screening workflows utilize ML surrogates to filter millions of candidates from existing databases based on stability, voltage, and migration barriers, successfully identifying promising cathodes for multivalent ions like calcium and magnesium [86] [14]. On the other hand, generative AI frameworks, including diffusion models and variational autoencoders, are being employed to design novel crystal structures and electrolyte formulations that do not exist in current databases. These generative approaches have shown success in discovering open-tunnel oxides for multivalent batteries and optimizing liquid electrolyte formulations through inverse design [225] [202]. However, benchmarking studies reveal a nuanced landscape: while generative models excel at proposing novel structural frameworks, baseline methods like ion exchange remain competitive for generating stable materials, suggesting a hybrid approach may be most effective [254].

Despite these advancements, critical challenges remain regarding data scarcity, model transferability, and the integration of experimental validation. The accuracy of ML models is often constrained by the quality and quantity of training data, leading to the development of transfer learning and continual learning strategies to mitigate catastrophic forgetting and improve performance in data-scarce regimes [217] [296]. The integration of robotic experimentation with differentiable modeling further bridges the gap between prediction and synthesis, enabling closed-loop optimization of electrolyte formulations [299]. As the field progresses, the synergy between physics-informed AI, high-performance computing, and automated experimentation promises to accelerate the discovery of next-generation solid-state battery materials, moving beyond incremental improvements to transformative breakthroughs in energy storage technology.

The Evolution of Data-Driven Workflows and Autonomous Systems

The landscape of materials discovery is undergoing a fundamental transformation, shifting from passive, single-shot property prediction to active, autonomous scientific reasoning. Historically, computational pipelines relied heavily on high-throughput density functional theory (DFT) screenings, which, while accurate, are computationally prohibitive for exploring the vast chemical space required for next-generation energy storage. Modern workflows are increasingly adopting automated systems that integrate multi-fidelity predictions, effectively bridging the gap between first-principles calculations and continuum device modeling. For instance, the AutoMat: Automated Materials Discovery for Electrochemical systems framework demonstrates how automated input generation and management of simulations across scales can seamlessly integrate machine learning surrogates or robotic experiments "in-the-loop," implicitly learning design features to optimize device performance without exhaustive DFT calculations for every iteration. This evolution addresses the bottleneck of computational resources, allowing researchers to navigate millions of candidates rapidly, as evidenced by pipelines that combined AI models with cloud high-performance computing to screen over 32 million candidates for solid-state electrolytes in less than 80 hours [6]. Such approaches have moved beyond simple filtering to actively guiding experimental validation, successfully identifying novel solid electrolyte compositions that were subsequently synthesized and characterized.

A critical advancement in this paradigm is the emergence of autonomous and multi-agent AI systems capable of executing the full discovery cycle. Unlike conventional machine learning approaches limited by the latent knowledge in their training data, these systems engage in hypothesis generation, experimental planning, and iterative refinement. The Autonomous Inorganic Materials Discovery via Multi-Agent Physics-Aware Scientific Reasoning ? study introduces SparksMatter, a multi-agent architecture that generates ideas, designs workflows, and critiques its own responses to propose novel stable inorganic structures that meet specific user objectives. Similarly, ChemNavigator: Agentic AI Discovery of Design Rules for Organic Photocatalysts employs agentic AI to autonomously derive structure-property relationships through hypothesis-driven exploration, successfully identifying statistically significant design rules for organic photocatalysts that previous ML approaches missed. These systems mark a transition toward active scientific reasoning, where AI does not merely predict properties but actively participates in the scientific method, suggesting rigorous follow-up validation steps including DFT calculations and experimental synthesis. This capability extends to battery-specific applications, where Expert-Guided LLM Reasoning for Battery Discovery: From AI-Driven Hypothesis to Synthesis and Characterization utilizes domain knowledge to steer LLM reasoning, successfully identifying, synthesizing, and characterizing novel cathode materials with significant capacity improvements.

Despite these advances, a significant challenge remains in the "synthesizability gap," where predicted stable structures often lack viable synthesis pathways. Addressing this requires a pivot to a synthesis-first paradigm, treating executable synthesis protocols as primary design variables rather than just atomic configurations [214]. Tools like ReactCA: A Cellular Automaton for Predicting Phase Evolution in Solid-State Reactions utilize cellular automaton simulations to predict phase evolution in solid-state reactions, offering a pathway to optimize recipes before experimental execution. Furthermore, the integration of large language models (LLMs) is revolutionizing how synthesis knowledge is extracted and utilized. Systems are now being developed to extract synthesis processes from scientific literature, creating structured corpora that inform computational experiment design [168]. This shift ensures that discovery pipelines prioritize materials that are not only thermodynamically stable but also experimentally accessible, closing the loop between computational prediction and laboratory realization.

The reliability of these data-driven workflows increasingly depends on the accuracy and efficiency of machine learning interatomic potentials (MLIPs). While universal potentials offer broad applicability, their performance in capturing high-energy landscapes and kinetic properties varies. Benchmarking studies reveal that while some models excel in static migration barrier predictions, others demonstrate superior fidelity in reproducing dynamic ion diffusivities, highlighting the need for careful model selection based on specific kinetic or thermodynamic requirements [66]. To address limitations in specific chemistries, such as the dynamic softness of halide electrolytes, domain-specific datasets like AQVolt26: High-Temperature r2SCAN Halide Dataset for Universal ML Potentials and Solid-State Batteries have been developed to co-train foundational models, ensuring robust performance under the elevated temperatures necessary for probing ion transport. Fine-tuning frameworks, such as An AI-Ready Fine-Tuning Framework for Accurate Machine-Learning Interatomic Potentials in Solid-Solid Battery Interfaces, further enhance accuracy for

complex interfaces like solid-solid battery boundaries, achieving quantum-level precision with significantly reduced data requirements. The convergence of these technologies-autonomous agents, synthesis-aware planning, advanced MLIPs, and generative design-is creating a self-reinforcing cycle of discovery that accelerates the identification of optimal materials and synthesis pathways for solid-state batteries.

Challenges in Data Scarcity, Heterogeneity, and Interface Characterization

The application of machine learning to battery materials discovery is fundamentally constrained by the scarcity of high-fidelity experimental and computational data, a limitation that is particularly acute for complex solid-state systems and buried interfaces. While traditional density functional theory provides accurate reference data, its prohibitive computational cost prevents the large-scale sampling required to train robust models for dynamic phenomena such as ion transport and interfacial degradation. To mitigate this data scarcity, researchers have increasingly turned to sophisticated data augmentation techniques and transfer learning strategies. For instance, lattice scaling has been proposed as an effective method to perturb unit cells isotropically or anisotropically, thereby expanding dataset sizes while respecting the physical volume changes inherent to insertion-based battery cycling [125]. Similarly, interpretable deep learning models utilizing transfer learning from lithium-ion datasets have successfully predicted electrode voltages for multivalent metal-ion batteries with significantly reduced error rates, overcoming the limitations of small dataset availability [217]. Furthermore, active learning frameworks that iteratively guide experimentation by leveraging both successful and unsuccessful results offer a pathway to alleviate data scarcity and mitigate human-induced biases in high-dimensional design spaces [141].

Despite these advances in data generation, the heterogeneity of battery materials necessitates the development of universal machine learning interatomic potentials capable of delivering near-DFT accuracy at a fraction of the computational cost. Recent benchmarking studies indicate that while various universal potentials show promise, their performance varies significantly across different material classes and properties. For solid-state electrolytes, models like MatterSim have demonstrated superior performance in predicting thermodynamic properties and lithium-ion diffusivity compared to other architectures, although discrepancies in force calculations remain a concern for some models due to insufficient training data coverage [171]. The reliability of these potentials for kinetic properties is also debated; while some models excel in static migration barrier predictions, others trained on high-temperature, non-equilibrium data demonstrate superior fidelity in reproducing dynamic ion diffusivities, suggesting that the inclusion of specific configurational sampling is critical for kinetic accuracy [66]. This is particularly relevant for soft, dynamically complex systems like halide electrolytes, where foundational models often fail to capture high-temperature distortions without co-training on domain-specific, off-equilibrium datasets [28]. Specialized pre-trained models, such as those designed for sulfide electrolytes with attention mechanisms, further illustrate that tailoring potentials to specific chemical families can yield remarkable accuracy in reproducing experimental ionic conductivity and stability across diverse doping scenarios [79]. Moreover, fine-tuning strategies that integrate efficient configurational sampling with replay-augmented continual learning have achieved quantum-level accuracy for solid-solid interfaces with only a fraction of the original dataset requirements [151].

Characterizing buried solid-solid interfaces and complex three-dimensional morphologies remains a formidable challenge that bulk property predictions cannot address. The "dirty" nature of battery surfaces, combined with the kinetic rather than thermodynamic governance of interfacial structures, complicates the creation of accurate simulation cells [282]. Traditional approaches struggle with the time-scale mismatch required to model ionic motion and relaxation in solid interfaces, necessitating the use of ML-driven molecular dynamics. Recent work utilizing fine-tuned ML potentials has unveiled detailed mechanisms of solid electrolyte interphase growth, such as the two-step reaction and crystallization process in argyrodite/Li-metal systems, which were previously inaccessible to conventional simulation methods [211]. Moreover, enhanced sampling techniques combined with deep potential molecular dynamics have revealed that lithium cluster nucleation often occurs within the interphase region rather than directly at the anode surface, driven by local electronic structure variations [24]. These atomistic insights are critical, as phase-field formulations coupled with viscoplastic descriptions of lithium deformation are now being used to predict void evolution and current hot spots that act as precursors for dendrite formation [198].

To bridge the gap between atomistic simulations and macroscopic performance, emerging techniques are combining ML with advanced imaging to create "digital twins" of battery microstructures. Generative adversarial networks integrated with stochastic geometry models allow for the calibration of 3D morphologies from 2D microscopy images, enabling virtual testing of all-solid-state battery cathodes under varied structural scenarios [130]. This approach facilitates the extraction of physical insights regarding triple phase junctions and conduction channels by transforming

multimodal X-ray images into topology-aware graphs [92]. Additionally, physics-guided continual learning frameworks are being developed to incrementally learn from new material data without catastrophic forgetting, ensuring that models remain robust as the chemical space of aqueous organic redox flow batteries and other systems expands [296]. These integrated approaches, ranging from data augmentation and universal potentials to digital twinning and continual learning, represent a paradigm shift toward overcoming the intrinsic data and interface challenges that have historically hindered the data-driven discovery of next-generation battery materials.

Generative Design, Inverse Modeling, and Hypothesis Generation

The paradigm of materials discovery is undergoing a fundamental transformation, shifting from traditional trial-and-error approaches and simple property prediction toward generative design and inverse modeling frameworks capable of navigating the vast, high-dimensional chemical space of solid-state materials. Generative artificial intelligence (AI) has emerged as a critical tool for proposing novel crystal lattices and complex molecular mixtures that transcend the limitations of human intuition. However, the efficacy of these generative models must be rigorously benchmarked against established baselines to ensure they offer genuine advantages in stability and novelty. Recent comparative studies indicate that while data-driven ion exchange methods remain superior for generating thermodynamically stable materials that closely resemble known compounds, generative techniques based on diffusion models, variational autoencoders, and large language models (LLMs) excel at proposing entirely novel structural frameworks and targeting specific properties such as electronic band gaps and bulk modulus [254]. To bridge the gap between theoretical generation and experimental viability, post-generation screening using universal interatomic potentials has become a standard practice, substantially improving success rates by filtering for stability and desired properties before synthesis attempts are made [254].

A significant advancement in this domain is the integration of symmetry-aware constraints and hierarchical architectures to manage the exponential complexity of crystal structure prediction. Traditional methods often struggle to locate stable regions within the vast search space of elemental combinations and three-dimensional arrangements. The Symmetry-aware Hierarchical Architecture for Flow-based Traversal (SHAFT) addresses this by decomposing the materials space into a hierarchy of subspaces defined by symmetric space groups, lattice parameters, and atomic positions. This approach significantly outperforms state-of-the-art iterative generative methods, such as Generative Flow Networks and Crystal Diffusion Variational AutoEncoders, by achieving higher validity, diversity, and stability in generated structures optimized for target properties [208]. Similarly, topology-constrained crystal structure prediction algorithms, such as TOPIC, have successfully identified promising oxide solid electrolytes by enforcing corner-sharing bond topology constraints, thereby reducing configurational complexity and enabling the discovery of high-performance candidates in previously unexplored chemical spaces [50]. These hierarchical and constraint-based strategies represent a move away from blind sampling toward guided exploration that respects the physical and chemical laws governing crystal formation.

Beyond structural generation, LLMs are revolutionizing the hypothesis generation phase of materials design by acting as force multipliers that integrate multidisciplinary knowledge beyond the cognitive limits of individual researchers. By leveraging materials system charts that encode processing-structure-property (P-S-P) relationships, LLMs can synthesize non-trivial design hypotheses involving synergistic interactions between mechanisms that are not explicitly present in the training literature. This capability has been demonstrated in the generation of innovative design ideas for high-entropy alloys with superior cryogenic properties and halide solid electrolytes with enhanced ionic conductivity, some of which have been experimentally validated in high-impact publications released after the models' knowledge cutoff dates [Beyond designer's knowledge: Generating materials design hypotheses via large language models]. Furthermore, agentic AI systems like ChemNavigator and SparksMatter extend this capability by autonomously deriving structure-property relationships and executing full discovery cycles, from ideation and planning to experimental workflow design and iterative refinement, consistently achieving higher scores in novelty and scientific rigor compared to frontier models [229; Autonomous Inorganic Materials Discovery via Multi-Agent Physics-Aware Scientific Reasoning]. These systems effectively mimic the scientific method, formulating hypotheses, designing experiments, and validating findings through rigorous statistical analysis, thereby establishing a framework for AI-assisted materials discovery that complements rather than replaces chemical intuition.

The application of generative and inverse design extends effectively to complex chemical mixtures and formulations, particularly for liquid and solid electrolytes where combinatorial spaces are immense. Unified frameworks integrating forward predictive models with inverse generative approaches have enabled the design of liquid electrolyte formulations with targeted ionic conductivity and solvation structures, marking a transition from single-molecule design to multi-condition constrained mixture generation [202]. In the realm of solid electrolytes, dual-stage machine learning frameworks combining fine-tuned machine learning interatomic potentials (MLIPs) with interpretable feature-property mapping have accelerated the discovery of high-entropy solid electrolytes, identifying compositions with high ionic

conductivity that traditional trial-and-error paradigms would likely miss [22]. Additionally, differentiable geometric deep learning models, such as DiffMix, unify physics-based mixture laws with data-driven approaches to guide robotic experimentation, successfully optimizing electrolyte ionic conductivity through gradient-based optimization in large chemical spaces [299].

Despite these advancements, challenges remain regarding the synthesizability of AI-generated materials and the accuracy of models in dynamic regimes. The current structure-centric paradigm often stalls at the "synthesizability gap," necessitating a pivot toward synthesis-first approaches where executable protocols are treated as primary design variables alongside atomic configurations [214]. This perspective argues that closing the gap demands representing synthesis procedures as machine-readable protocols and deploying generative models to propose actionable reaction pathways, integrated with closed-loop optimization to refine protocols against experimental realities. Moreover, while foundational MLIPs provide robust baselines, their accuracy can degrade in highly distorted, high-temperature regimes critical for probing ion transport in soft lattices like halides. Addressing this requires domain-specific configurational sampling and co-training with targeted datasets, such as AqVolt26, to ensure reliable dynamic screening and accurate prediction of ionic conductivities [28]. The convergence of generative models, hierarchical screening, and agentic reasoning represents a transformative trajectory, moving the field toward autonomous, closed-loop discovery systems capable of identifying chemically valid and physically meaningful materials at an unprecedented pace [150; 206].

Generative Architectures for Crystal Structure and Electrolyte Discovery

The paradigm of materials discovery has fundamentally shifted from the high-throughput screening of existing databases to the de novo design of novel crystal structures and complex chemical formulations, driven by advanced generative artificial intelligence. A primary obstacle in this domain is the vastness of the configurational space, where thermodynamically stable materials exist as isolated pockets within an exponentially large set of possibilities. To navigate this landscape efficiently, novel architectures such as the Symmetry-aware Hierarchical Architecture for Flow-based Traversal (SHAFT) have been developed. SHAFT decomposes the materials space into a hierarchy of subspaces consisting of symmetric space groups, lattice parameters, and atomic positions, allowing for the efficient exploitation of symmetry constraints. This hierarchical approach significantly outperforms iterative generative methods, such as standard Generative Flow Networks and Crystal Diffusion Variational Autoencoders, by achieving higher validity, diversity, and stability in generated structures optimized for target properties [208]. While these architectural innovations excel at proposing novel structural frameworks, comparative studies indicate that established baseline methods, such as data-driven ion exchange of known compounds, remain superior for generating stable materials that closely resemble existing prototypes. However, when sufficient training data is available, generative models demonstrate a distinct advantage in targeting specific properties like electronic band gaps and bulk modulus, particularly when augmented by low-cost post-generation screening using universal interatomic potentials [254].

In the specific context of energy storage, generative frameworks have proven instrumental in identifying host structures for multivalent ions, which are critical for next-generation batteries utilizing earth-abundant metals like aluminum and calcium. Traditional quantum-based models are often impractical for exploring the wide compositional range of open-tunnel oxides required for effective multivalent ion transport. By integrating Crystal Diffusion Variational Autoencoders (CDVAE) with fine-tuned Large Language Models (LLMs), researchers have successfully accelerated the discovery of stable transition metal oxide structures. These hybrid frameworks leverage machine learning to predict formation energies and energy above the hull, identifying candidates with lower formation energies than analogous compositions in existing databases, thereby suggesting enhanced thermodynamic stability for practical battery applications [225]. Similarly, for calcium battery cathodes, geometry-based descriptors combined with foundational machine-learned models have enabled the high-throughput screening of hundreds of thousands of structures to identify hosts with favorable migration barriers and voltage profiles [86]. The efficacy of these generative strategies is further highlighted in the discovery of inorganic electrides, where diffusion-based generation coupled with hierarchical screening has identified hundreds of new electron-rich compounds in previously unexplored chemical spaces [150].

Beyond solid crystal structures, the complexity of liquid and amorphous electrolytes necessitates unified frameworks that integrate forward predictive models with inverse generative approaches. Designing electrolyte formulations involves navigating a multi-objective landscape where properties such as ionic conductivity, solvation structure, and electrochemical stability must be balanced. Recent advancements have introduced the first generative machine learning frameworks specifically designed for molecular mixture design, capable of supporting multi-condition constrained generation. These systems integrate physics-informed architectures that preserve permutation invariance and incorporate empirical dependencies on temperature and concentration, enabling the experimental identification of liquid electrolytes with high ionic conductivity and specific anion-concentrated solvation structures [202]. Complementing this, chemical foundation models fine-tuned on extensive experimental datasets have been employed to discover novel high-conductivity formulations, achieving significant improvements in conductivity for specific salt systems through generative screening [159]. Furthermore, differentiable geometric deep learning models, such as DiffMix, extend mixture thermodynamic laws by creating learnable physical coefficients, allowing for gradient-based optimization of electrolyte properties in conjunction with robotic experimentation [299].

The success of these generative and inverse modeling strategies relies heavily on the accuracy and efficiency of the underlying property predictors, particularly machine learning interatomic potentials (MLIPs). The development of universal potentials trained on diverse datasets has enabled near-ab initio accuracy at a fraction of the computational cost, facilitating large-scale molecular dynamics simulations essential for understanding ion transport mechanisms in both crystalline and disordered systems. For instance, fine-tuned models have been critical in revealing the role of structural disorder in enhancing lithium-ion transport in solid electrolytes and in simulating the complex reaction dynamics at electrode-electrolyte interfaces [70] [187]. Moreover, the integration of these potentials with generative

models allows for the rapid relaxation and validation of generated structures, ensuring that proposed candidates are not only novel but also physically realizable. This synergy between generative design, inverse modeling, and high-fidelity simulation represents a transformative approach to materials discovery, enabling the systematic exploration of chemical spaces that were previously inaccessible to traditional experimental or computational methods. As these frameworks continue to evolve, incorporating active learning loops and multi-fidelity data, they promise to further accelerate the development of high-performance energy storage materials by bridging the gap between theoretical prediction and experimental realization.

Benchmarking Generative Models and Bridging the Synthesizability Gap

The integration of generative artificial intelligence into materials discovery has necessitated a rigorous re-evaluation of computational workflows, particularly concerning the thermodynamic stability and practical realizability of predicted compounds. While generative models such as diffusion networks, variational autoencoders, and large language models offer the allure of proposing novel structural frameworks, comparative benchmarks reveal that they do not universally outperform established data-driven methods. Specifically, traditional approaches like random enumeration of charge-balanced prototypes and data-driven ion exchange have demonstrated superior efficacy in generating thermodynamically stable inorganic crystals, even if the resulting candidates often closely resemble known compounds [254]. In contrast, generative techniques excel primarily when sufficient training data is available to target specific properties like electronic band gaps or bulk modulus, yet they frequently struggle to match the stability success rates of baseline methods without extensive post-generation filtering. To mitigate this disparity, recent workflows have incorporated low-cost screening steps utilizing pre-trained machine learning models and universal interatomic potentials, which substantially improve the success rates of both generative and baseline approaches by filtering for stability before expensive density functional theory (DFT) validation [254].

A pivotal evolution in addressing the limitations of structure-centric generation is the shift toward a "synthesis-first" paradigm. The prevailing focus on atomic configurations has created a significant "synthesizability gap," where thousands of candidate structures are proposed but few are experimentally realizable. Closing this gap requires treating executable synthesis protocols as primary design variables rather than secondary considerations [214]. This approach necessitates representing synthesis procedures as machine-readable protocols and deploying inverse-design models to propose actionable reaction pathways. Tools such as ReactCA, a cellular automaton framework, exemplify this direction by predicting phase evolution and intermediate formation in solid-state reactions based on precursor choice and heating profiles, thereby allowing for the *in silico* optimization of recipes prior to experimental execution [154]. Furthermore, thermodynamic selectivity metrics have been developed to assess the favorability of target phase formation against competing impurities, providing a data-driven foundation for predictive inorganic synthesis that moves beyond simple stability checks [284].

The challenge of bridging the gap between prediction and synthesis is further compounded by the complexity of chemical spaces in specific applications, such as electrolytes and high-entropy systems. In the domain of liquid electrolyte formulation, unified frameworks integrating forward predictive models with inverse generative approaches have proven effective in navigating the multi-objective nature of mixture design. These systems leverage physics-informed architectures to preserve permutation invariance and incorporate empirical dependencies, successfully identifying formulations with high ionic conductivity and specific solvation structures [202]. Similarly, chemical foundation models fine-tuned on extensive experimental datasets have enabled the discovery of novel high-conductivity electrolyte formulations, demonstrating that generalizable workflows can be adapted to tailor chemical mixtures for energy storage [159]. For solid-state systems, the discovery of high-entropy solid electrolytes has been accelerated by dual-stage machine learning frameworks that combine fine-tuned machine learning interatomic potentials (MLIPs) with interpretable feature-property mapping, overcoming the trial-and-error paradigms that previously constrained this high-dimensional composition space [22].

Despite these advancements, the reliability of universal machine learning interatomic potentials (uMLIPs) in capturing the high-energy landscapes required for kinetic modeling remains a subject of rigorous benchmarking. While models like Orb-v3 excel in static migration barrier predictions, others such as GRACE demonstrate superior fidelity in reproducing dynamic ion diffusivities, highlighting that architectural sophistication alone is insufficient without high-temperature, non-equilibrium training data [66]. This nuance is critical for synthesizability, as kinetic barriers often dictate whether a thermodynamically stable phase can actually be formed. Agentic AI systems are emerging to address these complexities by autonomously deriving design rules and executing full discovery cycles. For instance, systems like ChemNavigator and SparksMatter integrate large language model reasoning with physics-based calculations to formulate hypotheses, design experiments, and refine results iteratively, effectively mirroring the scientific method to derive chemically grounded design principles that complement human intuition [229] [Autonomous Inorganic Materials Discovery via Multi-Agent Physics-Aware Scientific Reasoning].

The transition from theoretical prediction to experimental validation is further supported by automated workflows that manage simulations across scales and integrate robotic experimentation. Frameworks like AutoMat accelerate the discovery pipeline by seamlessly integrating multi-fidelity predictions and automated input generation, while differentiable geometric deep learning models guide robotic platforms to optimize chemical mixtures, such as battery electrolytes, with remarkable efficiency [226] [299]. Moreover, the deployment of cloud high-performance computing alongside AI models has enabled the screening of tens of millions of candidates, leading to the experimental characterization of promising solid electrolytes and validating the potency of AI-guided experimentation [6]. Ultimately, the convergence of generative design, synthesis-aware protocols, and agentic reasoning represents a transformative trajectory in materials science, moving the field from the generation of hypothetical structures to the realization of practically synthesizable, high-performance materials.

High-Throughput Screening and Discovery of Novel Structural Frameworks

The exploration of new crystal structures is paramount for overcoming the intrinsic limitations of conventional electrolyte and electrode families, particularly as the field seeks to move beyond established frameworks like NASICON and garnets. Machine learning (ML)-accelerated high-throughput screening has emerged as a transformative methodology, enabling the rapid evaluation of thermodynamic stability and ionic mobility across thousands of hypothetical compounds. By utilizing geometric descriptors and foundational ML models, researchers can now identify viable host structures for both monovalent and divalent cations in previously uncharted chemical spaces. A critical advancement in this domain is the development of topology-constrained crystal structure prediction algorithms, such as TOPIC, which reduce configurational complexity by enforcing specific bond topology constraints like corner-sharing networks. This approach has successfully reproduced ground-state and metastable structures of known oxide solid-state electrolytes (SSEs) with large unit cells, leading to the identification of promising quaternary oxide candidates such as $\text{Li}_4\text{Hf}_2\text{Si}_3\text{O}_{12}$, which exhibits high ionic conductivity and stability [50]. Similarly, ab-initio random structure searching (AIRSS) has demonstrated the ability to locate low-energy structures of complex cathode materials based solely on chemical composition, predicting novel polymorphs of Fe-containing fluoroxalates that have yet to be experimentally synthesized [101].

While oxides offer chemical robustness, halide and sulfide electrolytes have gained attention for their enhanced ionic mobility and electrochemical stability. However, the dynamic softness of these lattices poses significant challenges for atomistic modeling, particularly under the elevated temperatures required to probe ion transport. The reliability of universal machine learning interatomic potentials (uMLIPs) in these regimes depends heavily on the quality and specificity of the training data. Recent work demonstrates that foundational datasets, while providing a strong baseline, often fail to capture the highly distorted, high-energy regions of the potential energy surface characteristic of soft halide and sulfide chemistries. The introduction of domain-specific datasets, such as AQVolt26, which includes high-temperature r2SCAN calculations for lithium halides, resolves these blind spots. Co-training with such targeted data significantly improves the accuracy of energy predictions and force transfers in distorted regimes, proving that domain-specific configurational sampling is essential for the reliable dynamic screening of halide electrolytes [28]. This finding is corroborated by studies on mixed lithium halides and sulfides, where fine-tuned universal potentials like PET-MAD, CHGNet, and DPA-SSE have been employed to disentangle the effects of lattice parameters and chemical composition on conductivity, revealing complex interplays between alloying, phase stability, and ion transport mechanisms [236] [104] [79].

The scope of high-throughput discovery extends beyond lithium systems to include sodium, magnesium, and calcium batteries, where the larger ionic radii and higher charge densities necessitate novel host frameworks. For sodium-ion batteries, large-scale data curation from multiple databases combined with descriptor-based ML models has enabled the prediction of average voltage and specific capacity for thousands of candidates, establishing a scalable framework for identifying stable, high-voltage cathode-active materials [19]. Similarly, geometry-based design principles, such as the use of Voronoi polyhedral volume as a descriptor for site compatibility, have facilitated the screening of over 50,000 structures to identify promising calcium cathode candidates with low migration barriers [86]. In the realm of magnesium-ion batteries, AI-driven workflows combining crystal graph convolutional neural networks with first-principles validation have successfully identified high-voltage topological quantum materials, demonstrating the transferability of ML models across different ion chemistries [14] [127]. Furthermore, hierarchical screening workflows have been developed to predict volume changes upon ion intercalation, allowing for the filtering of large candidate sets to accelerate the discovery of low volume change intercalation materials for batteries [139].

Generative artificial intelligence further expands the toolkit for materials discovery by proposing entirely novel structural frameworks rather than relying solely on known prototypes. Diffusion-based generative models and variational autoencoders have been benchmarked against traditional methods like ion exchange, showing superior capability in proposing novel structures when sufficient training data is available. When coupled with low-cost filtering steps using pre-trained interatomic potentials, these generative approaches significantly improve success rates in identifying thermodynamically stable materials [254]. This capability is particularly evident in the discovery of open-tunnel oxides for multivalent batteries, where generative AI frameworks integrating large language models and

crystal diffusion variational autoencoders have identified stable transition metal oxide structures with lower formation energies than analogous compositions in existing databases [225]. Furthermore, multiscale topological learning frameworks utilizing persistent homology have introduced new screening metrics, such as cycle density and minimum connectivity distance, to identify lithium superionic conductors by analyzing the connectivity of lithium-only and lithium-free substructures, leading to the discovery of novel candidates that resemble known superionic conductors in topological feature space [75].

Despite these advances, contradictions and challenges remain regarding the transferability of models and the accuracy of predictions across different material classes. While some studies suggest that stoichiometric descriptors alone can achieve competitive accuracy in predicting ionic conductivity, others emphasize the critical role of geometric and phonon-related descriptors in capturing the dynamic nature of ion transport [44] [48]. Additionally, the performance of universal potentials varies significantly; for instance, while some models excel in static migration barrier predictions, others demonstrate superior fidelity in reproducing dynamic diffusivities, highlighting that the inclusion of high-temperature, non-equilibrium training data is often the dominant driver of kinetic accuracy [66]. The integration of these diverse methodologies—from topology-constrained prediction and generative design to hierarchical screening with fine-tuned potentials—represents a paradigm shift in materials discovery, allowing for the systematic exploration of vast chemical spaces and the identification of next-generation battery materials that defy conventional chemical intuition.

Descriptor-Based Regression and Feature Selection for Conductivity

A pivotal advancement in materials informatics involves the systematic reduction of complex structural and compositional data into chemically intuitive descriptors, enabling the rapid identification of promising electrolyte candidates without the prohibitive cost of exhaustive first-principles molecular dynamics for every composition. This approach is particularly effective for NASICON-type electrolytes, where beta regression modeling has successfully isolated the primary modulators of diffusion. Research has demonstrated that the bottleneck width along diffusion paths and the average sodium-sodium distance are the dominant factors governing room-temperature self-diffusion coefficients [227]. Crucially, this study established that these complex geometric parameters could be accurately predicted using only two easily accessible descriptors: the sodium-ion content and the average ionic radius of the constituent metal ions. This reduction allows for the construction of explicit regression models that fit experimental data with high fidelity, facilitating the discovery of unexplored stable compositions with enhanced conductivity.

Parallel advancements in polymer electrolytes illustrate the versatility of sparse modeling techniques in extracting quantitative design rules from qualitative observations. In the realm of anion-conductive polymer membranes, Bayesian sparse modeling has been utilized to move beyond heuristic assessments of hydrophilic domain connectivity [235]. By calculating key constituent elements based on copolymer composition, this approach identified specific composition-derived features that serve as critical predictors for hydroxide ion conductivity. This shift from qualitative structural descriptions to quantitative, composition-based descriptors mirrors the success seen in inorganic systems, suggesting a universal utility for sparse modeling in deciphering the structure-property relationships of disordered or complex multiphase materials.

However, a significant tension exists between the interpretability of simple geometric descriptors and the necessity of capturing dynamic lattice effects. While studies on NASICON systems emphasize static structural parameters, other research argues that predictors restricted to static features may fail to account for the dynamic nature of ionic transport [48]. By curating features that include phonon-related descriptors derived from first-principles calculations alongside structural and electronic properties, recent work developed machine learning models that significantly improved prediction accuracy for solid electrolytes. These findings indicate that phonon-related features are essential for estimating ionic conductivity, suggesting that while static descriptors like bottleneck width are necessary, they may not be sufficient for capturing the full spectrum of transport mechanisms, particularly in systems where lattice dynamics play a pivotal role in ion hopping.

The debate over descriptor optimality extends to the relative importance of stoichiometric versus geometric features. Some investigations found that while global geometric descriptors improve tree-based predictions and enable interpretable structure-property analysis, stoichiometric descriptors, particularly the oxygen ratio, often dominate feature importance in gradient-boosted tree regressors [44]. This suggests that the optimal set of descriptors may be system-dependent, with some material families governed primarily by composition and others by specific structural motifs. Despite these nuances, the power of minimal descriptor sets is further evidenced by cross-domain predictions. A logistic regression-based model utilizing primarily theoretical elemental feature descriptors successfully distinguished between poor and good superionic conductors and achieved cross-domain classification, predicting lithium-based conductivity despite being trained exclusively on sodium-based examples [Machine learning-assisted cross-domain prediction of ionic conductivity in sodium and lithium-based superionic conductors using facile descriptors]. Through systematic permutation-based evaluation, this study reduced the feature space from 47 to just seven descriptors, reinforcing the argument that a minimal set of chemically intuitive features can often outperform complex, high-dimensional representations.

The integration of these descriptor-based approaches into high-throughput workflows has accelerated the discovery of novel frameworks. Platforms combining high-throughput screening with surrogate models have identified potential superionic conductors among thousands of lithium-containing materials by utilizing chemical descriptors and additional structural properties [1]. Furthermore, hybrid approaches have emerged to balance the speed of descriptor-based screening with the accuracy of dynamic simulations. For instance, integrating fine-tuned universal machine learning interatomic potentials with nudged elastic band calculations allows for the efficient prediction of migration barriers, addressing the limitations of static descriptors in capturing transition state energies [56]. While generative models and foundation models offer a new paradigm where descriptors are learned implicitly, the interpretability and chemical

insight provided by explicit descriptor-based regression remain indispensable for establishing fundamental design rules and guiding experimental synthesis.

Theory-Driven Exploration of Uncharted Chemical Spaces

While data-driven models have revolutionized materials discovery within known chemical domains, their efficacy is inherently constrained by the boundaries of existing structural databases, often failing to propose structures that deviate significantly from training distributions. Consequently, theory-driven approaches utilizing random structure searching (RSS) remain indispensable for mapping entirely new stable phases in ternary and multicomponent systems where prior knowledge is absent or misleading. A paradigmatic example of this capability is found in the exploration of the Ti-C-O and Al-C-O ternary systems, where researchers employed the ab initio random structure searching (AIRSS) technique coupled with complex network theory to map phase stability without recourse to structural databases [244]. This "data agnostic" approach successfully predicted the ambient pressure stability of novel carbonate compounds, specifically $\text{Ti}(\text{CO}_3)_2$ and $\text{Al}_2(\text{CO}_3)_3$, neither of which had been previously synthesized, underscoring the unique power of theory-driven methods to uncover chemically intuitive yet structurally unknown materials. Similarly, in the realm of cathode materials, AIRSS has demonstrated its utility by efficiently locating low-energy structures of complex Fe-containing polyanion compounds based solely on chemical composition, reproducing known polymorphs while predicting new, more stable variants that were hitherto unknown [101].

The necessity of such theory-driven exploration is further highlighted when contrasting these methods with massive-scale data-driven discovery efforts. Recent initiatives, such as the Graph Networks for Materials Exploration (GNoME) model, have identified nearly 400,000 thermodynamically stable crystalline materials, representing a significant acceleration in materials discovery [230]. However, a critical analysis of the GNoME dataset reveals that the vast majority of these stable materials involve elemental combinations containing rare or low-abundance elements, which casts doubt on their immediate practical application value [230]. Furthermore, while generative AI models excel at proposing novel structural frameworks, established baseline methods such as data-driven ion exchange often outperform them in generating materials that are both novel and thermodynamically stable, although these too tend to closely resemble known compounds [254]. This dichotomy suggests that while data-driven approaches can rapidly expand the known chemical space, they may struggle to access truly uncharted territories defined by unconventional stoichiometries or bonding environments without the guiding hand of physical theory.

To bridge the gap between the vastness of chemical space and the computational cost of first-principles calculations, recent advancements have integrated theory-driven search strategies with machine learning interatomic potentials (MLIPs). For instance, the discovery of oxide Li-conducting electrolytes in uncharted chemical spaces was achieved by introducing a topology-constrained crystal structure prediction algorithm, TOPIC, which enforces corner-sharing bond constraints to reduce configurational complexity [50]. This approach successfully identified promising candidates like $\text{Li}_4\text{Hf}_2\text{Si}_3\text{O}_{12}$ in quaternary oxide compositions, demonstrating that incorporating physical constraints into structure prediction can effectively navigate complex energy landscapes. Similarly, the prediction of stable Li-Sn compounds was significantly boosted by employing neural network potentials to accelerate ab initio searches, allowing for the screening of over a million structures and the identification of new ground states with large unit cells that would be prohibitive to find using density functional theory (DFT) alone [118]. These hybrid methodologies illustrate a synergistic trend where theory provides the search direction and physical constraints, while machine learning provides the computational efficiency required to explore high-dimensional configuration spaces.

Despite the successes of these hybrid approaches, challenges remain in ensuring the synthesizability and dynamic stability of predicted materials. The "synthesizability gap" remains a critical barrier, as many theoretically stable structures identified through high-throughput screening may not be accessible via conventional synthesis routes [214]. Addressing this requires a shift toward synthesis-first paradigms that treat executable protocols as primary design variables. Moreover, the accurate prediction of material properties often depends on capturing finite-temperature effects and dynamic behaviors, which static DFT calculations may miss. For example, machine learning-guided frameworks have been developed to predict temperature-induced solid-solid phase transitions by combining DFT with graph-based neural networks trained to estimate vibrational free energies, identifying thousands of potential transitions relevant for thermal management applications [178]. Additionally, the reliability of universal machine learning interatomic potentials in capturing high-energy landscapes relevant to ion kinetics is still being benchmarked, with studies showing that the inclusion of high-temperature, non-equilibrium training data is crucial for accurate dynamic transport predictions [66]. As the field progresses, the integration of rigorous theory-driven search algorithms with increasingly

sophisticated machine learning surrogates and synthesis-aware metrics will be essential for the reliable discovery of functional materials in the vast, unexplored regions of chemical space.

Machine Learning Interatomic Potentials for Atomic-Scale Transport Mechanisms

The transition from static barrier calculations to dynamic trajectory analysis has fundamentally altered the landscape of solid-state electrolyte research, primarily by overcoming the severe timescale and system-size limitations inherent to *ab initio* molecular dynamics (AIMD). Traditional AIMD is often restricted to small supercells and short durations, rendering it incapable of capturing rare diffusion events or statistically converging transport properties in complex, disordered systems. Machine-learned force fields (MLFFs) and deep potential models now bridge this gap, offering near-density functional theory (DFT) accuracy with computational speedups exceeding 350 times, thereby enabling the simulation of large supercells over hundreds of nanoseconds [49]. However, the fidelity of these simulations is contingent upon the quality of the training data; while universal machine learning interatomic potentials (uMLIPs) provide a robust baseline, they often struggle with the dynamically soft chemistries of halides at elevated temperatures unless augmented with domain-specific, high-temperature configurational sampling [28]. Recent benchmarking indicates that while architectural sophistication aids static predictions, the inclusion of non-equilibrium, high-temperature data is the dominant driver for accurately reproducing dynamic transport properties and structural correlations [66].

In the realm of amorphous solid electrolytes, MLFFs have elucidated microscopic transport mechanisms that appear insensitive to specific chemical compositions, a phenomenon difficult to rationalize without extensive statistical sampling. Large-scale trajectories have revealed that in amorphous oxyhalides, ionic conductivity is largely dictated by uncorrelated self-diffusion, evidenced by an alkali Haven ratio close to one, even though distinct-particle correlations significantly influence transference numbers [128]. This topological disorder creates a universal diffusion environment where local coordination, rather than elemental identity, governs transport, a finding further supported by studies on lithium phosphorous oxynitride (LiPON) where MLIPs revealed that transport anisotropy across the Li||LiPON interface is minor and unlikely to cause significant impedance buildup [173]. The capability to model such complex chemical spaces extends to disordered rocksalts and high-entropy systems, where conventional DFT screening is computationally infeasible, allowing for the exploration of highly disordered configurations that stabilize fast-ion conduction pathways [291][22].

For crystalline materials, ML-accelerated dynamics have uncovered nuanced transport mechanisms driven by the intricate coupling between lattice dynamics and ion mobility. In lithium zirconium chlorides, deep learning-accelerated simulations distinguished between collective diffusion in disordered phases and individual hopping in ordered phases, linking these behaviors to specific zirconium coordination environments and energy barriers [85]. Crucially, the role of anion dynamics has been quantified using constraint algorithms combined with MLMD, challenging the conventional view that facile anion rotation is the sole facilitator of superionicity; instead, anion translation, rotation, and vibration can each drive cation transport depending on the frequency match with cation hopping [122]. Furthermore, disorder-induced enhancements in lithium thiophosphates have been mapped using "softness" metrics derived from ML potentials, which classify specific hopping events and reveal how local structural fluctuations lower effective migration barriers [70]. These findings suggest that optimizing the dynamic coupling between the host lattice and mobile ions is as critical as minimizing static migration barriers.

Beyond bulk transport, ML potentials have revolutionized the study of interfacial phenomena and reactive processes by enabling explicit, reactive modeling of coupled mechanical, kinetic, and thermodynamic phenomena. In liquid electrolytes, Deep Potential-based MLMD has resolved early-stage solid-electrolyte interphase (SEI) nucleation, capturing spontaneous reduction reactions and bond rearrangements that classical force fields miss, revealing divergent growth pathways dependent on salt concentration and anion chemistry [40]. Similarly, large-scale reactive simulations of argyrodite electrolytes against lithium metal have unveiled a two-step growth mechanism involving amorphous phase formation followed by crystallization, supporting experimental hypotheses of indirect reduction pathways [211]. The ability to simulate millions of atoms over nanoseconds has also allowed for the identification of previously unreported crystalline disordered phases in SEIs, which evade purely thermodynamic predictions, underscoring the necessity of kinetic modeling to understand interface evolution [26]. Finally, the integration of MLFFs with Markov state models has established a quantitatively consistent link between atomistic hopping and mesoscale transport,

recovering rare diffusion processes that do not occur on AIMD timescales and validating that extended MLFF trajectories remove the strong finite-size bias present in shorter simulations [100].

Fine-Tuning Strategies and Benchmarking Universal Potentials

The paradigm for developing machine learning interatomic potentials (MLIPs) has decisively shifted from training models from scratch for every new material system to leveraging universal foundation models through strategic fine-tuning. This evolution addresses the prohibitive data inefficiency inherent in generating quantum-mechanical reference data, particularly for complex atomic-scale transport mechanisms where high-energy configurations are critical. A leading example of this approach is the Fine-tuning with Integrated Replay and Efficiency (FIRE) framework, which combines efficient configurational sampling with a replay-augmented continual learning strategy to adapt universal potentials to specific interfacial chemistries. By mitigating catastrophic forgetting—a common failure mode in continual learning where models lose fidelity to previously learned bulk properties—FIRE has achieved quantum-level accuracy in solid-solid battery interfaces, reporting root-mean-square errors in energy below 1 meV/atom and forces near 20 meV/Å while requiring only 10% of the original datasets [151]. Similarly, the concept of physics-guided continual learning (PGCL) has been extended to regression tasks in aqueous organic redox flow batteries, where task division strategies informed by sensitivity analysis optimize the learning process for emerging materials, ensuring that models retain robustness across diverse chemical spaces [296]. These frameworks underscore that the integration of physical priors and replay mechanisms is not merely beneficial but essential for maintaining model reliability when adapting to new interfacial environments.

Systematic benchmarking of universal MLIPs (uMLIPs) reveals a nuanced landscape where architectural sophistication often yields to the primacy of training data coverage. While many foundational models exhibit high fidelity in predicting static energies, significant deviations frequently arise in force calculations and dynamic transport properties if the training data lacks representation of specific high-energy configurations. For instance, benchmarking studies on alkali-ion battery kinetics indicate that while the Orb-v3 family excels in predicting static migration barriers due to architectural refinements, the GRACE model, trained on the OMat24 dataset, demonstrates superior accuracy in reproducing ion diffusivities and structural correlations [66]. This discrepancy highlights that the inclusion of high-temperature, non-equilibrium training data is the dominant driver of kinetic accuracy, surpassing the benefits of equivariant architectures alone. The necessity of such domain-specific data is further emphasized in studies of halide solid electrolytes, where foundational datasets provide a strong baseline for stable chemistries but fail to predict absolute energies accurately in distorted, high-temperature regimes essential for probing ion transport [28]. Co-training with targeted, high-temperature configurational data resolves these blind spots, suggesting that universal models must be augmented with system-specific sampling to serve as reliable zero-shot surrogates for dynamic screening.

The reliability of uMLIPs varies significantly depending on the specific transport mechanism and the chemical environment, necessitating rigorous validation before deployment. In the context of solid-state electrolytes, benchmarks comparing models such as MatterSim, MACE, CHGNet, and SevenNet reveal that MatterSim often achieves the best agreement with reference values for lithium-ion diffusivity, particularly at room temperature, whereas other models may exhibit deviations due to insufficient training data coverage or energy consistency issues [171]. Conversely, for reaction dynamics involving bond breaking, such as the decomposition of ethylene carbonate at lithium metal interfaces, fine-tuning foundation models like MACE-MP0 on ab initio data generated via umbrella sampling is essential to obtain statistically converged free energy profiles and accurate reaction rates [187]. Without such fine-tuning, standard density functionals or unrefined potentials can overestimate reaction rates by orders of magnitude, leading to erroneous mechanistic conclusions. Furthermore, the performance of graph-based MLIPs in modeling disordered systems, such as lithium-based disordered rocksalts, shows that while some potentials like AENET excel in energy predictions, others like Moment Tensor Potentials (MTP) are superior for atomic forces, indicating that no single architecture currently dominates across all metrics [291].

The interplay between model architecture and data quality is also evident in the study of high-entropy solid electrolytes and complex cathode materials, where fine-tuned potentials enable high-throughput discovery workflows. A dual-stage machine learning framework combining fine-tuned CHGNet potentials with interpretable feature-property mapping has successfully identified high-conductivity compositions in the quinary HE-LZSP space, demonstrating that fine-tuned potentials can balance computational efficiency with generalization reliability [22]. However, the transferability of these models remains a challenge; while universal potentials perform well on familiar compositions, their predictive power degrades when applied to novel chemical spaces without additional fine-tuning on relevant local environments.

This limitation is addressed by workflows that integrate active learning to guide targeted experimentation, systematically leveraging both successful and unsuccessful results to refine models incrementally [141]. Moreover, the use of fine-tuned potentials in nudged elastic band (NEB) calculations has enabled high-throughput screening of migration barriers, identifying promising fast-ion conductors that would be computationally prohibitive to assess using density functional theory alone [56]. These collective findings establish that while universal potentials provide a robust starting point, rigorous validation and strategic fine-tuning with domain-specific data are indispensable prerequisites for deploying MLIPs in the predictive simulation of atomic-scale transport mechanisms.

Unraveling Ion Transport Mechanisms in Disordered and Amorphous Systems

The application of machine learning-driven molecular dynamics (ML-MD) has fundamentally reshaped the understanding of ion transport in disordered and amorphous solid-state electrolytes, revealing dynamic phenomena that static models and traditional heuristic design rules fail to capture. In amorphous oxychloride systems, specifically the Li-Al-O-Cl family, simulations have elucidated a unique structural mechanism where oxygen doping induces the formation of a skeleton composed of Al-chains formed by AlOCl tetrahedra. This structural reorganization enhances the glass-forming ability and facilitates the rotation of chlorine atoms around the central aluminum within the tetrahedra, which in turn promotes the motion of lithium ions [263]. However, this enhancement is non-monotonic; an excessive increase in the oxygen-to-chlorine ratio reduces the population of rotating chlorine atoms, thereby weakening lithium transport. This finding suggests that optimizing glass-forming ability without compromising the halide content is critical for maximizing conductivity in these systems. These specific insights align with broader universal design principles established through atomistic simulations of 32 different Li-M-X systems, which identify that saturated M-X groups with more negative charges preferentially form M-X-M chains, and that the bulk modulus serves as a simple yet effective predictor for Li⁺ mobility [17].

Similar principles of structural disorder enhancing transport are observed in lithium thiophosphates, where deep learning potentials demonstrate that disorder-driven diffusion dynamics significantly boost room-temperature conductivity [70]. By employing a machine learning-based structure fingerprint known as "softness," researchers have successfully classified disorder-induced "soft" hopping lithium ions, establishing a quantitative bridge between local structural features, atomic rearrangements, and macroscopic dynamical characteristics. Building on this, new theoretical frameworks utilizing ML-based softness analysis have introduced the concept of "path entropy" to explain anomalous fast ion conduction in disordered materials. Traditional design heuristics often rely on reducing activation enthalpy to promote ionic diffusion, yet this approach is frequently constrained by the enthalpy-entropy compensation mechanism, or the Meyer-Neldel Rule. Certain disordered solid-state electrolytes defy this rule, suggesting an elusive source of entropy governs their kinetics. The proposed path entropy metric accounts for strong fluctuations in the free energy landscape perceived by lithium cations, enabling an assessment of collective diffusion behavior that transcends static, composition-based entropy approaches [58]. This unified framework connects structural disorder directly to the freedom of ionic diffusion pathways, offering a new dimension for designing superionic conductors.

The role of anion dynamics in facilitating cation mobility has also been re-evaluated through the lens of ML-MD, challenging the prevailing view that facile anion rotation is the primary driver of cation transport. Recent studies combining constraint algorithms with machine learning molecular dynamics reveal that anion translation, rotation, and vibration can each directly drive superionicity, with one mode typically dominating depending on the material class [122]. Anion rotation dominates when its frequency matches the cation hopping frequency, whereas anion translation prevails at higher frequencies and vibration at lower ones. These anion motion modes fundamentally drive cation migration by inducing large disorder within the cation sublattice and altering local cation dynamics, effectively lowering energy barriers and affecting the prefactor of diffusion. In amorphous oxyhalides, this interplay is further complicated by the formation of chain fragments of metal-anion tetrahedra. Large-scale simulations indicate that while oxygen anions on these tetrahedra can limit alkali diffusion, the overall transport is largely dictated by uncorrelated self-diffusion, as evidenced by an alkali Haven ratio close to one [128]. This behavior stands in stark contrast to crystalline conductors, where distinct-particle correlations often play a more significant role, and explains why the diffusion properties of amorphous oxyhalides remain largely insensitive to variations in atomic chemistry.

In specific halide systems like Li₂ZrCl₆, deep learning-accelerated simulations have uncovered distinct transport mechanisms dependent on the crystalline phase and degree of disorder. The disordered alpha-phase exhibits the highest ionic conductivity, driven by collective diffusion and anisotropic interlayer transport, whereas the ordered beta-phase shows significantly lower conductivity dominated by isotropic translations and intralayer migration [85]. Furthermore, in cation-disordered Li₃TiCl₆, ML-MD simulations have captured anti-correlated Li-ion motion, a phenomenon rarely reported in solid-state materials. Analysis of the self and distinct parts of Li-ion dynamics reveals that transport occurs mainly through interstitial hopping, with the anti-correlation suggesting complex many-body interactions that are critical for accurate conductivity predictions [55]. Collectively, these ML-driven insights challenge traditional static

views of ion transport, emphasizing the critical roles of dynamic anion coupling, structural disorder, and entropic contributions in facilitating rapid ion conduction in complex, non-crystalline environments.

Conclusion

The literature reviewed herein delineates a profound paradigm shift in solid-state battery research, moving from static, trial-and-error experimentation to a dynamic, data-driven ecosystem powered by artificial intelligence and autonomous workflows. This transformation is anchored in the convergence of high-fidelity curated datasets and advanced machine learning interatomic potentials, which have fundamentally altered the scale and speed of materials discovery. These tools now enable the rapid screening of vast chemical spaces and the accurate simulation of ion transport mechanisms that were previously inaccessible due to computational constraints. The transition from simple descriptor-based regression to sophisticated graph neural networks and universal potentials has allowed researchers to capture the complex, non-linear relationships governing ionic conductivity, thermodynamic stability, and electrochemical performance across diverse chemistries, including emerging sodium, calcium, and magnesium systems.

Despite these advancements, significant challenges remain in bridging the gap between theoretical prediction and experimental realization. A critical tension exists between the universality of foundational models and the necessity for domain-specific, high-temperature training data to accurately capture the kinetic behavior of dynamically soft materials. Furthermore, the field is increasingly recognizing that thermodynamic stability alone is insufficient for predicting battery performance; kinetic pathways, interfacial reactivity, and mechanical degradation mechanisms such as void evolution and dendrite growth play decisive roles. The emergence of concepts like interfacial breathing and reactive memory underscores the need for multiscale modeling frameworks that integrate atomistic reaction dynamics with mesoscale chemo-mechanical evolution to fully understand failure modes in all-solid-state batteries.

Looking forward, the trajectory of battery materials discovery points toward the realization of fully autonomous, self-driving laboratories that close the loop between hypothesis generation, synthesis, and characterization. The integration of generative AI for inverse design, agentic systems for scientific reasoning, and robotic platforms for execution promises to overcome the synthesizability gap by treating synthesis protocols as primary design variables. However, the success of this integrated ecosystem depends on resolving current contradictions regarding model transferability, addressing data scarcity in multivalent systems through interpretable transfer learning, and establishing standardized benchmarks for kinetic accuracy. By harmonizing physics-informed AI with rigorous experimental validation, the community is poised to accelerate the development of next-generation energy storage solutions that are not only theoretically optimal but also practically viable and commercially scalable.